Machine Learning Project

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1 Analyses using Unsupervised Learning

1.1 Introduction

The given dataset in this section is the European Working Conditions Survey 2016. This survey was carried out based on ages and genders as the demographic variables. The objectives of the analyses in this section are to visualise and discover some underlying patterns in the dataset. The results and information pertaining to the European working conditions can be potentially useful for the authorities and agencies in Europe for implementing necessary actions or enhancements. Unsupervised learning methods will be solely considered in this section, primarily Principal Components Analysis (PCA).

1.2 Principal Components Analysis (PCA)

In PCA, we would like to reduce the dimensionality of the variables and also to find the representative factors that can be the best explained in the data. A principal component that defines a direction in the variable space associated with the largest variance is first calculated. This first principal component Z1 with its loading vectors is subsequently represented by a linear combination of the original variables. Then, the subsequent principal components Zi with their respective linear combination are calculated based on maximal variance with no correlation to previous principal components.

```
> # Loading vectors of principal components based on variables
> pca.out$rotation
             PC1        PC2          PC3         PC4           PC5         PC6          PC7         PC8         PC9         PC10         PC11
Q2a  0.03203956 -0.1386327  0.796373784  0.57638908 -6.171166e-02  0.01266193 -0.092896333  0.01060833 -0.02186704  0.005570217 -0.009937025
Q2b  0.07652230 -0.2204528 -0.584133839  0.76073419  7.105450e-02  0.00515712  0.005490225 -0.12077633  0.01372336  0.070150927  0.028533515
Q87a 0.39103574 -0.1996019 -0.038763673 -0.07849823 -3.148653e-02  0.02786038 -0.179630910 -0.07249015 -0.62888270 -0.261821916 -0.544290070
Q87b 0.37759153 -0.2359578  0.077079602 -0.16741716 -4.488656e-02  0.08133873  0.158131884 -0.36972267 -0.28416331  0.574242537  0.432370395
Q87c 0.39652146 -0.2056496 -0.004550283 -0.03679735 -1.796326e-02  0.05172394  0.097909301  0.16394260  0.07090937 -0.668278083  0.554994648
Q87d 0.37141006 -0.2534245  0.062704331 -0.09378305 -5.747547e-05  0.14878121  0.360503139 -0.19974893  0.62699603  0.013748398 -0.446981468
Q87e 0.36263461 -0.1259478 -0.059239797 -0.08174241 -3.299479e-02 -0.14466435 -0.712223779  0.37711846  0.29943899  0.284447504  0.019249723
Q90a 0.33784962  0.3007859  0.002609147  0.12630062  1.210966e-01 -0.20735048  0.495518337  0.62792742 -0.15661941  0.226314662 -0.078670047
Q90b 0.27485090  0.4436706  0.054692725  0.05645151  2.715430e-01 -0.62004729 -0.100309447 -0.47561321  0.09433706 -0.131861974  0.026155642
Q90c 0.22363116  0.5038874  0.015633656  0.08498139  3.729994e-01  0.71576928 -0.177731352 -0.06749933  0.01374979  0.001025211  0.028835352
Q90f 0.17680118  0.4160141 -0.080175825  0.10806045 -8.713190e-01  0.08308652 -0.008177733 -0.09643027  0.04070890 -0.020991855 -0.002154017
```

Figure 1.1

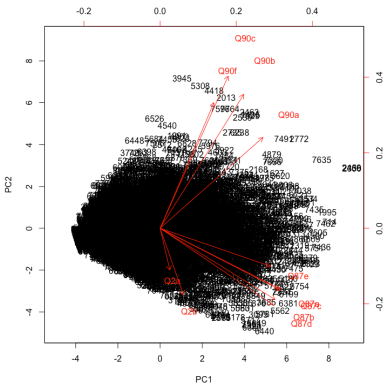

Figure 1.2

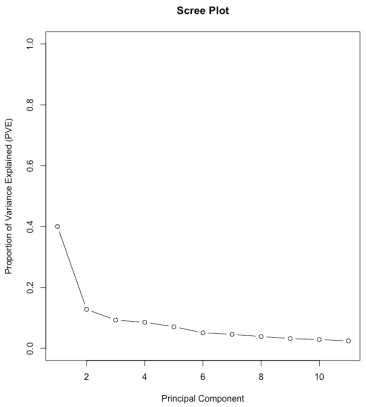

Figure 1.3

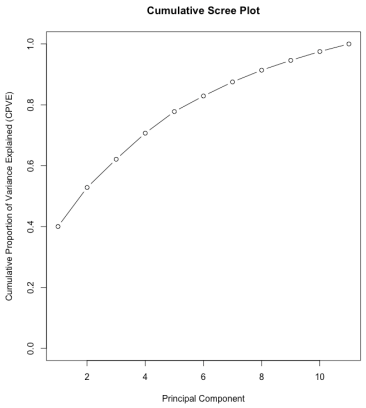

Figure 1.4

By eyeballing the scree plot in figure 1.3 and looking for an “elbow” in the plot where a point at which the proportion of variance explained (PVE) drops off, the best amount to choose for principal components is two. However, the first two principal components are merely explaining 52.85% of the variance which can be clearly seen from figure 1.4 which the cumulative scree plot shows the cumulative proportion of variance explained. I recommend choosing the first three principal components as they have accounted for 62.1% of the variance explained. From the social science context, the proportion of variance explained more than 60% would be fine. This is also supported by the scree plot where the plot suggests us to choose three principal components. The first three principal component factor loadings can be obtained in figure 1.1 which are PC1, PC2 and PC3.

2 Regression Problem

2.1 Introduction

Education is playing an extremely important role in a secondary school's student life. Some of them like to score high marks and probably there is a group of students who are not that interested in learning. Here we are told to predict the final scores of students for the subjects Mathematics and Portuguese Languages as they have a fairly high fail rate. As they also contribute to the success rate of school's subjects, we would like to build a regression model using the variables involved to better understand the variables and factors that affect student performance in both of these subjects. The final predicted results collected will then benefit decision makers to improve on the quality of education in the two schools, Gabriel Pereira and Mousinho da Silveira.

2.2 Aims of an optimal regression model and technique

As we aim to optimise the model, we exclude the first (G1) and second period grade (G2) in training the regression models for final period grade (G3) as three of them are highly correlated. Thus, we aim to minimise the multicollinearity among predictors as much as possible. We will predict the final period scores of Mathematics and Portuguese language respectively. Here we also aim to improve the model interpretability and its predictive performance. We will be using the root mean squared error (RMSE) to examine the regression models. RMSE is used because it tells us the standard deviation between a predicted and the corresponding actual value. Hence, we would like to lower the RMSE by comparing between different models. The multiple linear regression started off with 30 variables (excluding G1, G2, G3) in predicting final period scores (G3 - between 0 to 20) for both Mathematics and Portuguese language.

Apart from that, the data will have a train-test split ratio of 0.7 for the train set. The training set is used to train the two models in which they are then implemented on the test set. This technique allows us to analyse the predictive performances of the corresponding models.

2.3 Multiple Linear Regression

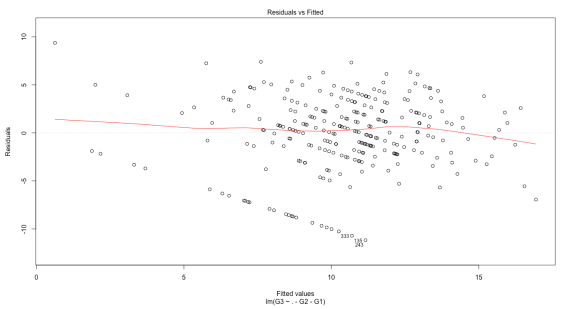

Figure 2.1 (Maths)

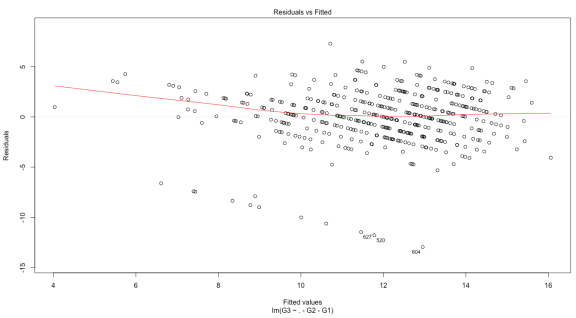

Figure 2.2 (Portu)

Figures 2.1 and 2.2 show the difference between the actual G3 scores and the predicted G3 scores using the multiple regression model. As we can clearly observe that the residuals plot of Mathematics and Portuguese language in the figure 2.1 and 2.2 do not obviously have any non-linear patterns shown as it seems the residuals are equally spread around the horizontal red fitted line. Hence, the non-linear models will not be considered in these regression problems.

```
> rmse(summ_m) # for math
[1] 3.701912
> rmse(summ_p) # for portuguese
[1] 2.597313
```

Figure 2.3 from R

We can observe that the root mean squared error between the actual and predicted final score G3 of Mathematics and Portuguese Language are 3.7 and 2.6 respectively. We then seek to lower the RMSE if it is possible.

2.4 Tree-based Model (CART: Classification and Regression Tree)

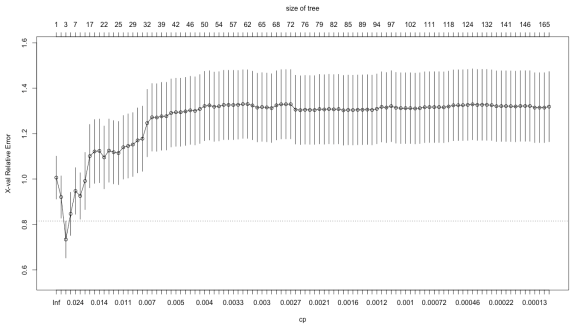

Figure 2.4 (Maths)

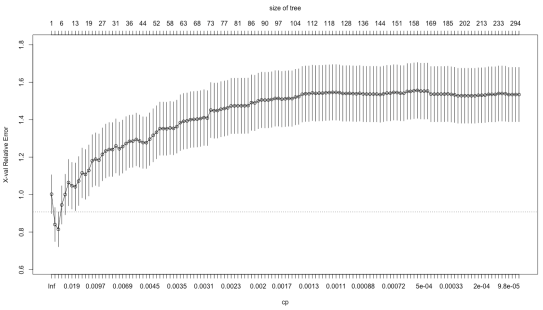

Figure 2.5 (Portu)

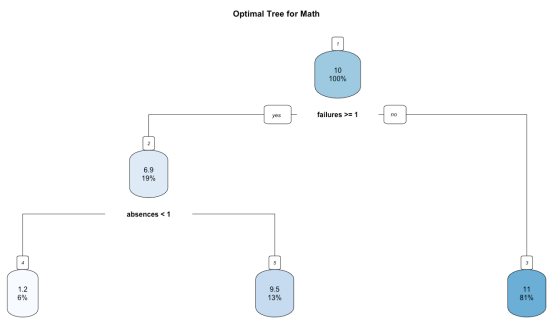

Figure 2.6 (Maths)

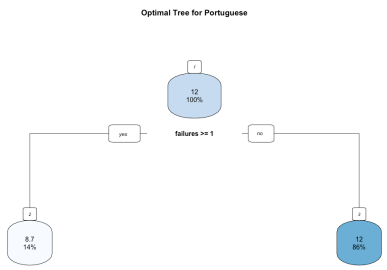

Figure 2.7 (Portu)

```
> cart_m_prune$variable.importance
 absences  failures       age     goout  guardian    higher
797.46352 787.96171 154.50230 143.73057  92.70138  77.25115
```

```
> cart_p_prune$variable.importance
failures      age
784.8880  48.3008
```

Figure 2.8 (Maths)

```
sqrt(mean((G3hat_m - G3.test_m)^2))
.] 4.344429
```

Figure 2.9 (Portu)

```
> mean((G3hat_p - G3.test_p)^2) %>% sqrt()
[1] 2.762415
```

Figure 2.10 (Maths)

Figure 2.11 (Portu)

We trained a tree-based model for this regression problem, also known as CART. Here we trained regression trees (CART) for the comparison to the previous model, multiple regression. There will be a set of splitting rules which is used to divide the predictor space into different regions. It has one identifier of splitting the regression tree which is using the variance reduction. We would like to choose the biggest variance reduction for the splitting rule. After splitting the observations into the child node with the larger variance reduction from the best splitting rule, we repeat the calculation of the variance reduction till there is no difference between observations. In the regression tree model, the prediction of new observations will be based on the mean in its region and then it will be predicted as the mean of that region. In this CART model, the tree is first grown to the maximum and is then pruned based on the cross-validated error using the 1-SE rule in R. The pruned tree is simple, stable, and easy for visualisation and interpretation based on the importance of the variables.

I have selected 3 subtrees in ‘maths’ data and 2 subtrees in the ‘portu’ data as it has the lowest cross validation error as shown in the figure 2.4/2.5 respectively. The pruned subtree in Figure 2.6 (Mathematics) depicts that among students who had number of past class failures (failures) more than or equals 1, those whose number of school absences (absences) less than 1 had an average score of 1.2, and if otherwise, they had an average score of 9.5. Students who had failures less than 1 had an average score of 11. The pruned subtree in Figure 2.7 (Portuguese) depicts that students who had failures more than or equal 1 had an average score of 8.7 and those who had failures less than 1 had an average score of 12.

In figures 2.8 and 2.9, CART assigns a higher variable importance score in the pruned tree for the primary splitters, meaning that it is the level of important variables in the regression tree. In figure 2.10/2.11, the root mean square error of this regression tree model on the test data is 4.34 and 2.76 for Mathematics and Portuguese respectively.

2.5 Conclusion

	RMSE (Mathematics)	RMSE (Portuguese)

Multiple Linear Regression	3.701912	2.597313
Regression Tree	4.344429	2.762415

Table 2.12

In table 2.12, the multiple linear regression is still the better model for its high predictive performance which has the fairly lower root mean square error for both the similar data of predicting G3 scores.

3 Classification Problem

3.1 Introduction

Nowadays, the bank would be more likely to do marketing for their term deposit. Therefore, we would like to build a classification model to predict whether a client will subscribe to a term deposit of the bank or not and also to make the marketing campaigns more efficient. We plan to build a classification model to better understand the important variables involved in predicting client subscription to the term deposit. The model can play an vital role in helping the person-in charge of the marketing campaign better manage their resources by prioritising higher potential subscription clients.

3.2 Aims of an optimal classification model and technique

To make the model more optimal, we cannot include the variable which is not suitable when making predictions in the future with the unseen data. Thus, the variable ‘duration’ is taken out in our classification analyses.

To find the better model among all the classification models, we zoomed out our aims to improve the model interpretability and the predictive performance and chose the two simple classification algorithms. To predict the qualitative dependent variable (yes/no) which is the term deposit in this case, we chose two classification models which include the logistic regression and the tree-based model (Classification and Regression Tree - CART). Logistic regression is the basic and fairly strong model for classification problems as it uses the logistic function [p(X) = odds / (1+odds)]. Classification trees look more practical for decision making and are relatively easy for visualisation and interpretation. We need the pruned tree to reduce the maximum size of tree for the decision making. It can be used for better decision making for the marketing campaign of term deposit in this case. Classification accuracy is used as a metric to compare between the two different models which are logistic regression and classification tree.

Besides that, the data will have a train-test split ratio of 0.664. This is because we have 4521 rows of data, (4521x0.664) is approximately equal to 3000, and therefore can perform 10-fold cross validation in the r.part library in R. The default fold of cross validation in the r.part library in R is 10 folds. The training set is used to train the two models in which they are then implemented on the test set. This technique allows us to analyse the predictive

performances of the corresponding models.

3.3 Logistic Regression

```
> coef(LR.bank)
       (Intercept)               age     jobblue-collar    jobentrepreneur       jobhousemaid      jobmanagement         jobretired
     -1.541415e+00      1.541418e-03      -3.241936e-01      -3.335515e-01       1.497670e-01      -8.015608e-02       8.195712e-01
  jobself-employed        jobservices         jobstudent      jobtechnician      jobunemployed         jobunknown     maritalmarried
      2.371591e-01     -1.157526e-01       2.991064e-02      -3.389117e-01      -2.060343e-01       7.020610e-01      -6.771877e-01
     maritalsingle educationsecondary  educationtertiary  educationunknown         defaultyes            balance         housingyes
     -3.711467e-01      1.885369e-01       3.750219e-01      -2.859032e-01       5.413880e-01      -1.799526e-05      -4.725898e-02
           loanyes   contacttelephone     contactunknown                day           monthaug           monthdec           monthfeb
     -7.570894e-01      8.047442e-02      -1.013837e+00       1.327751e-02      -2.660939e-01       6.836034e-01       1.468302e-01
          monthjan           monthjul            monthjun           monthmar           monthmay           monthnov           monthoct
     -1.235681e+00     -7.304115e-01       5.759796e-01       3.899570e-01      -5.635378e-01      -7.717926e-01       1.478920e+00
          monthsep           campaign              pdays           previous      poutcomeother    poutcomesuccess    poutcomeunknown
      5.172933e-01     -8.012326e-02       6.866840e-04       4.696097e-02       1.639081e-01       2.258824e+00       1.687775e-01
```

Figure 3.1

As we mentioned above, logistic regression is the basic and fairly strong model for classification problems as it uses the logistic function [p(X) = odds / (1+odds)]. Thus, in this case, it models the probability of a term deposit subscription by using the logistic function. The odds can be calculated using the exponentiated coefficients of the predictors in the figure 3.1. The logistic function is almost the same as the sigmoid function. Therefore, if the value of odds close to 0, meaning that it has a very low probabilities of client subscription to the term deposit, vice versa. This means that the logistic function yields a probability close to one for clients who subscribed and a probability close to zero for clients who did not subscribe. In this logistic regression model, I have made an assumption that the threshold is 0.5 when calculating the probability, meaning that the probabilities p(X) > 0.5 predict term deposit subscription as “yes”, otherwise it predicts term deposit subscription as “no”. However, banks can select a higher threshold to reduce the number of false positives. It sounds good as the bank no need to arrange their resources to entertain for the falsely predicted term deposit subscription client where it is “yes”.

```
"Classification accuracy = 0.8855"  "Classification error = 0.1145"
```

```
                y.test
LR.bank.predict   no  yes
             no 1319  156
            yes   18   27
```

Figure 3.2 Figure 3.3 Figure 3.4

The classification accuracy of this logistic regression model on the test data is 0.8855. The classification error is 1- classification accuracy and thus it is 0.1145. The classification accuracy can be calculated using the confusion matrix as shown in the figure 3.4. Sum of true positive and true negative followed by dividing with the sum of all numbers in figure 3.4. (1319+27) / (1319+156+18+27). This model is good enough but we seek for another better model to have a more accurate prediction using unseen data in the future.

3.4 Tree-based Model (CART: Classification and Regression Tree)

Here we trained classification trees (CART) for the comparison with the previous model, logistic regression. There will be a set of splitting rules which is used to divide the predictor space into different regions. There are two identifiers of splitting the decision tree in this classification context which are using the entropy and gini index. The more the entropy removed, the bigger the information gain. After splitting the observations into the child node with the largest number of information gain from the best splitting rule, we repeat the calculation of the information gain till all the same class of observations are almost in the same leaf. In the classification tree model, the prediction of new observations will be based on the mode in its region and then it will belong to that class. In this CART model, the tree is first grown to the maximum and is then pruned based on the cross-validated error using the 1-SE rule in R. The pruned tree is simple, stable, and easy for making decisions based on the importance of the variables.

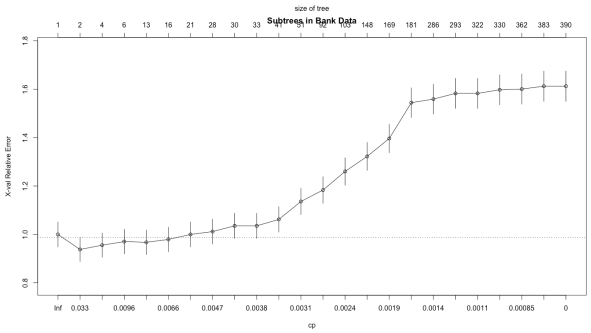

Figure 3.5

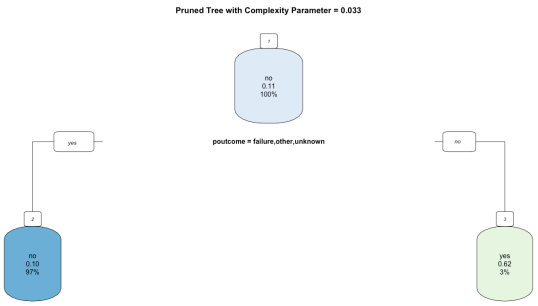

Figure 3.6

I have selected 2 subtrees in bank data as it has the lowest cross validation error as shown in the figure 3.5. The pruned subtree in Figure 3.6 depicts that if the outcome of the previous marketing campaign (poutcome) for a client was unsuccessful (failure, other, unknown), the client is predicted to not subscribe a term deposit, and it is predicted to subscribe a term deposit otherwise.

```
> prune.bank$variable.importance
poutcome
45.66671
```

Figure 3.7

	y.test	
tree.pred	no	yes
no	1323	153
yes	14	30

Figure 3.8

```
"Classification accuracy = 0.8901"
```

Figure 3.9

In figure 3.7, CART only has a variable importance score in the pruned tree, meaning that it is the most important variable in the classification tree and can further exploit this variable by making decisions for the improvement in marketing campaign. In figure 3.9, the classification accuracy of this classification tree model on the test data is 0.8901. The classification accuracy can be calculated using the confusion matrix as shown in the figure 3.8. (1323+30) / (1323+156+14+30).

3.5 Conclusion

The CART has a slightly higher classification accuracy which is 0.8901 compared to logistics regression which is only 0.8855. We will then choose CART as a classification model for predicting the term deposit subscription of clients. Also, we strongly recommend using the CART model as it gives a better model interpretability through a series of if-else conditions. The CART model allows the marketing team of the bank to easily identify the potential customers and therefore leads the marketing team to be more successful in hitting their goals with higher efficiency.